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To cite this article: Eliaber Barros Santos, Ariane Morgana Leal Soares, Bismark Lopes Bahia, Rosane Rodrigues Da Costa Pereira & Carla Fernanda Fávaro (2022) First record of *Zelus pedestris* (Hemiptera: Reduviidae) preying on the cacao pest *Monalonia bondari* (Hemiptera: Miridae), Biocontrol Science and Technology, 32:4, 511-514, DOI: [10.1080/09583157.2021.1990857](https://doi.org/10.1080/09583157.2021.1990857)

To link to this article: <https://doi.org/10.1080/09583157.2021.1990857>



Published online: 20 Oct 2021.



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BRIEF REPORT



First record of *Zelus pedestris* (Hemiptera: Reduviidae) preying on the cacao pest *Monalonion bondari* (Hemiptera: Miridae)

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ABSTRACT

Monalonion bondari is an important *Theobroma cacao* pest in Brazil and information about their natural predators is lacking. Thus, the aim of this study was to record for the first time the predation of *M. bondari* by *Zelus pedestris* in the field and confirm this predatory activity in the laboratory. In the laboratory, predation by *Z. pedestris* resulted in the mortality of 45% of nymphs and 40% of adults of *M. bondari*. This is the first record of *Z. pedestris* preying on *M. bondari* and can be a potential species to be used for the biological control of this pest.

ARTICLE HISTORY

Received 12 July 2021

Accepted 1 October 2021

KEYWORDS

Assassin bug; biocontrol;
Theobroma cacao

Monalonion bondari Costa Lima, 1938 (Hemiptera: Miridae) is the main pest of cacao plants, *Theobroma cacao* Linnaeus., in Brazil (Sánchez, 2011); *Monalonion* species have been recorded in 13 Neotropical countries (Gamboa et al., 2020), and some of these countries are important cocoa producers. The entire life cycle of this *M. bondari* occurs on cacao plants, and it is responsible for major losses of cacao crops.

Females of *M. bondari* deposit endophytic eggs in cocoa pods and nymphs of all stages and newly emerged adults can be found in developing fruits. The terminal branches of the plant attacked by adults exhibit necrosis and wilting of the plant tissue followed by death. In places where *M. bondari* is uncontrolled, complete crop losses have been recorded (Maia & Valverde, 2017).

The control of *M. bondari* in Brazil is exclusively chemical, and in Bolivia, the use of *Beauveria bassiana* under experimental conditions has yielded satisfactory results for the control of *M. dissimulatum* (Ferrari et al., 2014). The entomopathogenic fungi and the use of insect predators such as *Podisus* sp. (Heteroptera: Pentatomidae), *Heza* sp. (Heteroptera: Reduviidae) (Bustos & Reyes, 1973), and *Azteca chartiflex spiriti* Forel, 1912 (Hymenoptera: Formicidae) (Bondar, 1937) are promising for use in cacao integrated pest management. The purpose of this study was to report predation of *M. bondari* by *Zelus pedestris* Fabricius, 1803 in the field and confirm this predatory activity in the laboratory.

While monitoring cacao plants in the Vale do Juliana farm, municipality of Igrapiuna, Bahia, Brazil (13°51'16.9"S, 39°17'02.2"W), in September 2020 (spring season in Brazil), a

fifth instar bug nymph was found preying on a third instar nymph of *M. bondari*. The predator was taken to the laboratory and noted to be a female. The species was identified as *Zelus pedestris* Fabricius (1803) (Hemiptera: Reduviidae) by Dr. Hécio Reinaldo Gil Santana, from the Oswaldo Cruz Foundation (FIOCRUZ – Rio de Janeiro, Brazil) based on Zhang et al. (2016). After identification, additional specimens were collected, and some groups were established in transparent cylindrical plastic cages (17 cm height $\times$ 14 cm diameter) in the laboratory.

To confirm the predatory activity of *Z. pedestris* on *M. bondari* in the laboratory, we followed a similar methodology to that of Speranza et al. (2014). The insects were kept at $24 \pm 3^\circ\text{C}$, with $70 \pm 20\%$ relative humidity and a natural photoperiod (12 h light:dark cycle). Individuals of *M. bondari* were confined in transparent cylindrical plastic cages (17 cm height $\times$ 14 cm diameter) screened with voile fabric, together with *Z. pedestris* and cacao fruit used as a food source for *M. bondari*. Two treatments were established with four replicates each: (1) one nymph of *Z. pedestris* (instar V) and five nymphs of *M. bondari* (instar IV/V), and (2) one adult female of *Z. pedestris* and 5 adults of *M. bondari* (males/females).

In the field, we observed *Z. pedestris* eggs and nymphs on cacao leaves and fruits in areas infested by *M. bondari* (Figure 1). In the laboratory tests, the number of predated individuals was quantified after 48 h, and predatory activity was observed from adults and nymphs of *Z. pedestris* (Figure 2). A total of 40% of adults and 45% of nymphs of *M. bondari* were preyed upon by *Z. pedestris*. The first predation of *M. bondari* adults was registered after eight minutes from the beginning of observation. *Z. pedestris* consumed the entire haemolymph of *M. bondari* and the result of this this was exuvia-shaped remains, except for adults who maintained their body shape, but were completely dehydrated.



Figure 1. Eggs of *Zelus pedestris* found in a cacao-growing area infested by *Monalonion bondari*. (a) oviposition in fruit; (b) oviposition in the abaxial region of a cacao leaf.

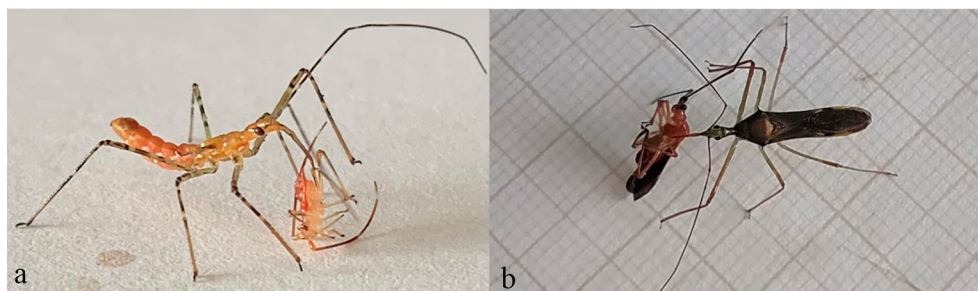


Figure 2. Predation of *Monalonion bondari*. (a) a nymph of *Zelus pedestris* (V instar) preying on a nymph of *M. bondari* (IV instar); (b) a female of *Z. pedestris* preying on a female of *M. bondari*.

We observed the presence of *Z. pedestris* in the three different cacao cultivation systems, in sun-grown cacao plants in Mutuípe ($13^{\circ}20'48.5''\text{S}$, $39^{\circ}32'30.5''\text{W}$), in an agroforestry system with *Hevea brasiliensis* in Maraú ($14^{\circ}14'22.9''\text{S}$, $39^{\circ}05'45.1''\text{W}$), and further south in an agroforestry system with native species from the original forest (cabruca) in Ilhéus ($14^{\circ}47'44.6''\text{S}$, $39^{\circ}10'19.2''\text{W}$). The presence of *Z. pedestris* eggs in the whole cacao plant in areas infested by *M. bondari* shows the potential of this species as a biological control agent for the management of this much-feared pest.

Predators belonging to the Reduviidae family are voracious and generalists and this condition increases their ability to succeed both for the control of *M. bondari* (Hemiptera), as well as for several other cacao pests belonging to other orders such as Lepidoptera, Thysanoptera, Coleoptera, and Hymenoptera (Maia & Valverde, 2017). Generalist predators often attack multiple species, and this favours the establishment of populations of these natural enemies, since they do not suffer from the lack of specific food. Therefore, generalist predators can be good agents for biological pest control. Insects belonging to the genus *Zelus* are a good example since they have been reported to feed on caterpillars, beetles, and stink bugs in wheat crops (Gassen, 1986), stink bugs in South America and adjacent islands of the Caribbean (Zhang et al., 2016), and on *Perkinsiella saccharicida* (Kirkaldy, 1903) (Hemiptera: Delphacidae) on sugarcane in Ecuador (Mora et al., 2004).

To the best of our knowledge, this is the first record of *Z. pedestris* preying on *M. bondari* and further studies are needed to obtain more insights about the biological aspects and role of this generalist predator in *M. bondari* and other cacao pests.

Acknowledgements

We would like to thank Dr Hécio Reinaldo Gil Santana, at Oswaldo Cruz Foundation (FIOCRUZ – Rio de Janeiro, Brazil) for the identification of the species. We are also grateful to the partners of Fazenda Vale do Juliana for the opportunity to monitor the cacao cultivation areas. Moreover, we would like to thank M & A Burton for the suggestions and English editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by CAPES [grant number 001].

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